a recently recovered host. In addition to culture techniques, serological techniques are also used for diagnosis of diseases.

Usefulness of Koch's Postulates

- It is useful in determining pathogenic organisms.
- To differentiate the pathogenic and nonpathogenic microorganism.
- For the classification of organisms.
- To detect the susceptibility or resistance of the laboratory animals.

1.2.5 Joseph Lister(1827-1912)

Joseph Lister was a British surgeon (Figure 1.8). He found out that microorganisms were responsible for wound infections. He developed a system of antiseptic surgery. He used bandages soaked in phenol solution to prevent wound infection. He sterilized instruments by heat and sprayed diluted phenol over surgical area and prevented contamination of wounds. He was the first person to isolate bacteria in pure culture using liquid culture. Thus, he was considered as co-founder of Medical Microbiology with Koch, who later isolated bacteria on solid media.

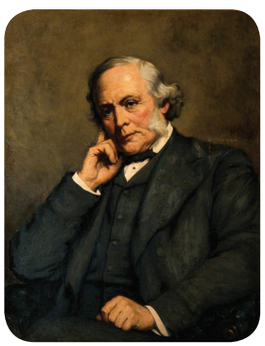


Figure 1.8: Joseph Lister (1827-1912)

1.2.6 Alexander Fleming (1881-1955)

He was a British Bacteriologist. He observed a mold (*Penicillium notatum*) growing on a plate of *Staphylococcus aureus*. The growth of *Staphylococcus aureus* around the mold colony was inhibited (Figure 1.9). He also showed that the culture filtrate of mold inhibited the growth of *Staphylococcus aureus*. He called this substance Penicillin, which acted on Gram positive bacteria. For the discovery of this antibiotic Fleming (Figure 1.10), Florey and Chain got Nobel Prize in 1945. Penicillin eventually came into use during world war II as a result of the work of a team of scientists led by Howard Florey of the University of Oxford.

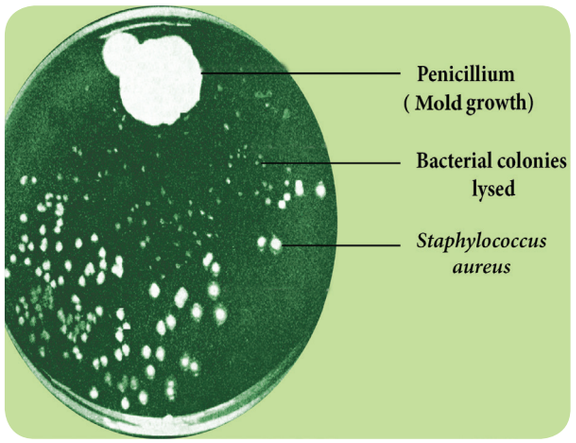


Figure 1.9: Original culture plate on which the observation of action of penicillin was made by Alexander Fleming



Figure 1.10: Alexander Fleming (1881-1955)



Alexander Fleming, the discoverer of penicillin warned about the possibility of antibiotic resistant bacteria due to antibiotics misuse, as early as in 1920s.

TN_GOVT_XI_Micro_Biology_CH01.indd 8 16-02-2019 9.54.23 AM